BankShield AI *Generative AI Powered Platform for Automated Reverse Engineering, Malware Analysis and Risk Scoring of Fraudulent Mobile Applications*

Problem Statement

Fraudulent mobile applications targeting Indian banking customers are increasing rapidly. These apps impersonate legitimate banking applications, steal credentials, intercept OTPs, abuse accessibility services, and perform unauthorized financial transactions. Manual APK analysis is time-consuming and requires significant cybersecurity expertise that most banks lack internally.

Solution Approach

BankShield AI is an intelligent cloud platform that performs automated analysis of Android APK files and generates comprehensive security risk assessments for banking institutions and cybersecurity teams.

Module 1 — APK Upload and Processing Analysts or banks upload suspicious APK files via a secure web interface. The system extracts the APK contents including manifest, DEX bytecode, certificates, assets, and embedded resources for analysis.

Module 2 — Static Analysis Engine The engine extracts and analyzes:

- Declared permissions (especially dangerous permissions: SEND_SMS, READ_CONTACTS, BIND_ACCESSIBILITY_SERVICE)
- Activities, Services, Receivers, and Content Providers
- Hardcoded URLs, IP addresses, and API endpoints
- Signing certificates and developer identity
- Suspicious API calls (reflection, dynamic code loading, native code execution)
- Banking keyword patterns and typosquatting indicators

Module 3 — Dynamic Behavior Analysis Engine The APK is executed in an isolated Android sandbox environment to observe runtime behaviors including:

- Network communications and C2 server connections
- Accessibility service abuse and overlay attacks
- SMS interception and OTP harvesting attempts
- Credential harvesting via fake login screens
- Unauthorized financial transaction attempts
- Anti-analysis and evasion techniques

Module 4 — AI-Based Malware Classification Machine learning models trained on labeled Android malware datasets analyze the extracted static and dynamic features to classify applications into four risk categories: Benign, Suspicious, High Risk, and Malware. Feature vectors include permission combinations, API call sequences, network behavior patterns, and code structure signatures.

Module 5 — Generative AI Explanation Engine Large Language Models (LLMs) convert raw technical findings into human-readable security reports. The GenAI engine explains detected malicious behaviors in plain language, identifies attack techniques using MITRE ATT&CK Mobile framework mappings, and generates actionable mitigation recommendations for banking security teams.

Module 6 — Risk Scoring Dashboard A web-based dashboard presents risk scores (0-100), threat classifications, detailed findings, and AI-generated reports. Banks can track analyzed APKs, monitor threat trends, and share intelligence across institutions.

System Architecture

APK Upload → Static Analysis → Dynamic Sandbox → ML Classification → GenAI Reporting → Risk Dashboard

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Datasets and Data Sources

Dataset	Description	Size
CICMalDroid 2020	Android malware samples across 5 categories	17,341 samples
CIC-AndMal2017	Labeled benign and malicious APKs	10,854 samples
Drebin Dataset	Android malware with feature vectors	5,560 malware samples
AndroZoo Repository	Large-scale APK collection with VirusTotal labels	20,000+ samples
MalwareBazaar	Active malware threat intelligence feed	Real-time
OpenPhish	Phishing URL intelligence	Real-time
CERT-In Advisories	India-specific security advisories	Public

These datasets provide labeled benign and malicious Android applications for model training, validation, and evaluation. Real-time feeds ensure the system stays current with emerging threats.

Technology Stack

Component	Technology
Backend	Python, FastAPI
Static Analysis	Androguard, Apktool
Dynamic Sandbox	Android Emulator, Frida
ML Classification	scikit-learn, XGBoost

Component	Technology
GenAI Engine	Amazon Bedrock (Claude)
Database	PostgreSQL, Redis
Frontend	React, Tailwind CSS
Deployment	AWS (EC2, S3, Bedrock)

Innovation

The key innovation is the integration of Generative AI with traditional malware analysis to automatically explain malicious behaviors in plain language. Unlike existing tools that output raw technical data, BankShield AI generates actionable security intelligence accessible to banking officials without deep cybersecurity expertise.

The platform connects directly with AnteClick, a consumer-facing phishing detection app, creating a two-layer defense: BankShield AI protects banks by analyzing threats at the institutional level, while AnteClick protects individual users in real time on their devices.

Expected Outcomes

- Reduce APK malware analysis time from hours to minutes
- Improve fraud detection capability for Indian public sector banks
- Provide explainable AI-generated security reports accessible to non-technical banking staff
- Support rapid identification of fraudulent banking applications targeting Indian customers
- Enable real-time fraud intelligence sharing across Public Sector Banks in India
- Reduce financial losses from mobile banking fraud

Team

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